

# Nikhil Mittal

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## EXPERIENCE

### Software Development Engineer Intern

February 2026 – Present

*SwiffyLabs*

*Bengaluru, Karnataka*

- Replaced custom counter tables and pessimistic locking with PostgreSQL sequences and optimistic inserts, improving ID generation throughput 4x sequential and 2.5x random under load tests.
- Evolved a versioned ClickHouse query engine from flat AND-only filters to an AST-based query model supporting nested boolean expressions while preserving backward compatibility with existing clients.
- Integrated Maker-Checker, Authentication, and Policy Engine services for policy-driven authorization via **gRPC**.
- Optimized PostgreSQL indexes and SQL queries, eliminating sequential scans for OTP configuration lookups.

### Research Program Participant

October 2025 – January 2026

*Samsung PRISM Program*

*Remote*

- Built an LLM-based categorization pipeline for SmartThings automations, processing **50+ JSON samples**.
- Authored prompt templates and parsing logic to enforce consistent category assignment and title accuracy.

### Full Stack Developer Intern

June 2025 – August 2025

*Centre of Excellence in Data Science and AI (TIET-UQ), TIET*

*Patiala, Punjab*

- Engineered backend for the first LLM-powered system prototype at the lab, supporting **2 role-based dashboards**.
- Established centralized service layers handling authentication, user workflows, and inference orchestration.
- Decoupled LLM inference modules from core services to improve modularity, scalability, and deployment velocity.

### Frontend Developer Intern

June 2025 – August 2025

*Cohyve*

*Remote (Part Time)*

- Refactored and enhanced **3 React modules** integrated with Node.js and MongoDB backend APIs in production.
- Diagnosed and resolved production issues across 2 critical user workflows, improving reliability for **100+ users**.

## PROJECTS

### Aayu Mitra Elderly Voice Assistant [\[GitHub\]](#) | Python, OpenAI API, Firestore, LLMs, STT/TTS

- Architected backend integrating LLM APIs with **STT**, **TTS**, and context to power an AI elderly care assistant.
- Implemented **Firestore** session persistence and device identity mapping for stateful, multi-session conversations.
- Designed multilingual prompt pipelines with fallback logic for reliable LLM-based emotion and intent support.

### Wired Collaborative Platform [\[Live\]](#) | React, Node.js, MongoDB, Selenium, JWT, REST APIs

- Developed **10+ REST APIs** with JWT authentication and role-based access control for collaboration workflows.
- Modeled MongoDB schemas and aggregation pipelines powering mentor matching for a **1,500+ student** users.
- Delivered end-to-end request lifecycle logic ensuring data consistency and integrity across users and projects.

## EDUCATION

### Thapar Institute of Engineering and Technology

September 2022 – June 2026

*B.E. Computer Engineering | CGPA: 8.58/10.0*

*Patiala, Punjab*

## TECHNICAL SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, SQL, C++

**Backend:** Spring, Spring Boot, Microservices, REST APIs, gRPC, API Design, Node.js, Express, FastAPI, Auth

**Databases:** PostgreSQL, ClickHouse, MongoDB, Redis, MySQL, Firestore

**Tools:** Git, Azure DevOps, Docker, Postman, ghz

**GenAI & AI Tooling:** LLM Orchestration, Prompt Engineering, OpenAI API, STT/TTS Pipelines

**Core CS:** Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks

## POSITIONS OF RESPONSIBILITY

### Technical Coordinator

November 2022 – May 2024

*Microsoft Learn Student Chapter, TIET*

*Patiala, Punjab*

- Led technical planning and execution of a **500+ participant** Makeathon, increasing participation by **25%**.
- Launched and deployed a 30 Days of Code platform using Firebase and GitHub APIs with live progress tracking.
- Mentored **30+ members** on web development and system design through structured technical sessions.